Beyond the Surface

Evaluating Perception and Accuracy in 3D Printed Data Visualizations

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Overview

- Motivation and Background
- Roadmap
- Projects
  1. 3D Bar Charts
  2. Experiential Learning
  3. 3D Heat Maps
- Contribution

Experiential learning serves a dual purpose by letting Stat 218 students interact with the study, thus letting us get a large sample size

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Motivation

Motivation

Modern statistical graphics exist largely within the confines of 2D projections.

- Research articles
- Textbooks
- Documentation / reports
- News articles

Why?

- Easy to make / understand
- Low cost to produce
- Many available software options

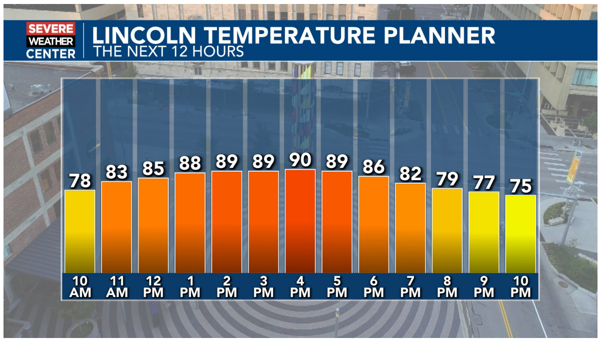

Figure 1: *Source: www.1011now.com*

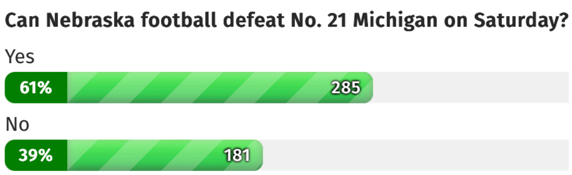

Figure 2: *Source: www.klkntv.com*